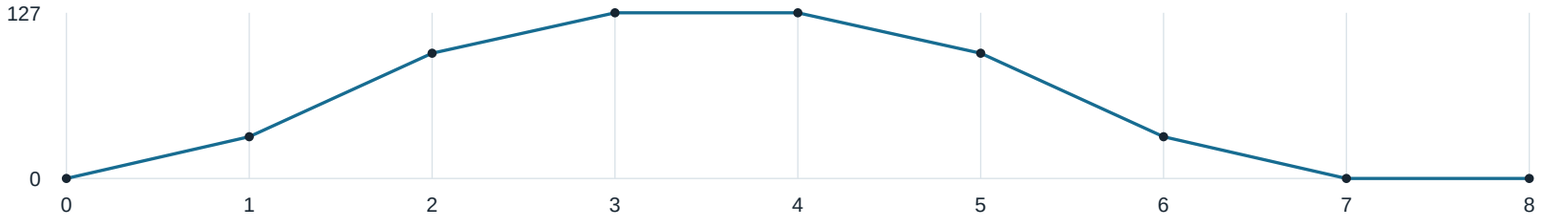
MIDI CC smoothing

Same nine CC74 values. Equal beat spacing for a direct shape comparison.
Dorico 5 uses sampled straight segments; the exact curve is a reference here.

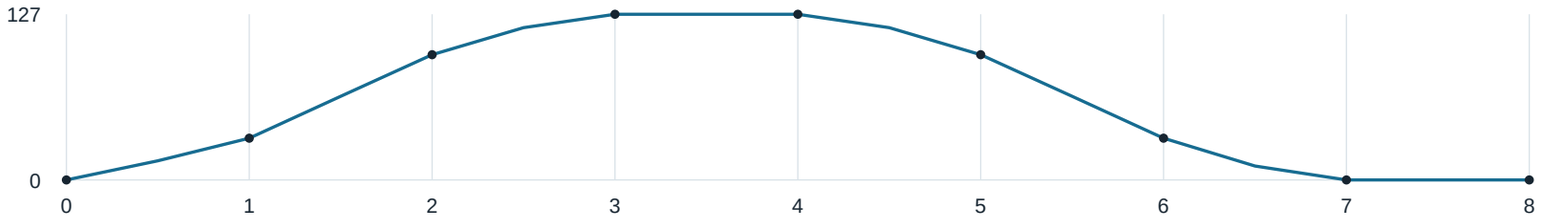
1 Straight ramps between source events

Eight segments; corners remain at the MIDI events.



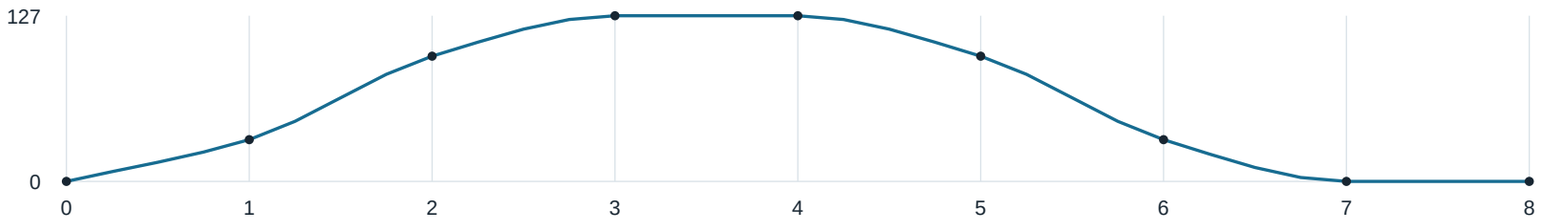
2 Previous smoothed contour - 16 segments

Eighth-note sampling of the cubic model.



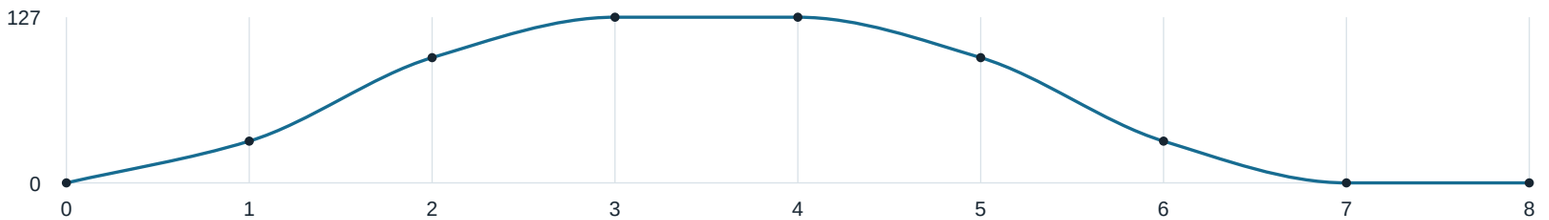
3 New smoothed contour - 32 segments

Sixteenth-note sampling; generated and tested as native Dorico lines.



4 Exact cubic Bezier reference

Eight cubic spans with matching tangents; this row is not a Dorico object.



Estimated maximum error from the cubic model: 2.345 CC (16) -> 0.775 CC (32).
About 3x closer with twice the objects. The comparison uses musical time, not page distance.

ControllerNotation | 6 November 2026 | Black dots: source MIDI values | X: quarter-note beats